Next-Technique Prediction in an APT29 Evaluation Sequence: A Pilot Benchmark

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***Abstract*—Forecasting an adversary's next action is a useful goal for threat-informed defense, but ordered public attack traces are scarce. This paper reports a pilot benchmark of next-technique prediction on one ordered sequence reconstructed from the MITRE ATT&CK Evaluations APT29 material. The five vendor result files yield identical ordered substep/technique pairs, but they describe the same emulation flow and are not independent campaigns. Candidates are drawn from an independent, version-pinned ATT&CK Enterprise v18.1 catalogue of 835 technique IDs, not from the evaluation sequence. Models are trained on 500 heuristic tactic-constrained synthetic sequences per seed plus the first 98 observed steps, and are scored by teacher-forced top-k ranking on the remaining 42 steps. Over three seeds, mean top-5 accuracy is 12.7% for first-order Markov, 11.9% for second-order Markov with backoff, 6.4% for a local co-occurrence ranking, 5.6% for an LSTM, and 1.6% for global frequency; uniform random ranking would give 0.6%. Three additional generator-policy checks show that scores also depend on tactic sampling and whether revoked/deprecated techniques are allowed in synthetic training: co-occurrence top-5 ranges from 3.2% to 11.9%, while LSTM top-5 ranges from 0% to 5.6% across tested conditions. These are descriptive pilot results on 42 sequentially dependent positions from one emulation; they do not establish cross-campaign or operational prediction capability. An earlier 58-label experiment whose scores cannot be reproduced is reported separately and is not comparable.**

***Index Terms*—MITRE ATT&CK, adversarial technique prediction, sequence modeling, Markov model, LSTM, cybersecurity evaluation, synthetic data**

I. Introduction

MITRE ATT&CK provides a shared vocabulary for adversary tactics and techniques and is widely used in threat intelligence, detection engineering, and incident response [1]. A more ambitious task is predicting which technique may follow a partially observed sequence. Such predictions could help analysts prioritize investigation, but they are not deterministic forecasts: tactics express adversary goals, and operations can branch, repeat, omit, or reorder behaviors [2].

Prior ATT&CK research has studied technique associations and co-occurrence across reported groups and malware [3], [4]. Association is not equivalent to temporal predictability, and research on telemetry mapping and report extraction addresses different tasks [5]–[8]. Execution-ordered public ground truth remains a practical constraint on directly evaluating next-step prediction.

This paper makes three limited contributions:

- It defines next-technique prediction as a top-k ranking task over an independent, versioned candidate universe (ATT&CK Enterprise v18.1, 835 IDs), with fail-closed coverage checks so that no evaluation label can be added to the candidate set.
- It reports a fully specified, deterministic, audited run of five ranking methods on a 98/42 chronological split of one APT29 sequence, with every ranked list released and every hit count recomputed independently.
- It documents why the numbers cannot support cross-campaign or operational claims, including a quantified sensitivity to the synthetic generator's boundary rule.

II. Related Work

A. ATT&CK Association and Sequence Modeling

Al-Shaer *et al.* studied statistical associations among techniques and used clustering to infer unobserved techniques from observed ones [3]; Erfan *et al.* studied co-occurrence in adversary and malware technique data [4]. Both motivate association-based baselines while leaving open whether ordered prefixes help predict the next technique.

A systematization of ATT&CK research identifies application areas and validation gaps [1]; a later review covers 417 peer-reviewed publications [5]. Mapping low-level telemetry to ATT&CK [6] and extracting techniques from reports with language models [7], [8] are related but different from one-step-ahead forecasting.

Sequential threat forecasting predates technique-level work. Ghafir *et al.* combine alert correlation with a hidden Markov model to predict APT stages [9]. Pappu studies smoothed higher-order Markov models for tactic prediction [10]. Their prediction units and protocols differ from ours, so their metrics are not comparable.

Most directly, Raj *et al.* introduce a hybrid LSTM–Markov framework for attack-chain forecasting using 4,849 campaign chains, and their publisher notes that the severity analysis has 33 independent campaign-level labels [11]. This paper therefore does not claim to be the first study of ATT&CK sequence forecasting; its scope is a top-k comparison on one APT29 evaluation sequence. Metrics are not comparable across the two studies, and no claim of advantage over the hybrid approach is made.

B. Synthetic Attack Data and Emulation Sources

APTGen generates artificial targeted-attack sequences from incident reports and executes corresponding activity [12]. The present generator is much simpler: a tactic-constrained random walk without report grounding or execution. The real sequence is reconstructed from public APT29 evaluation results [13],

[14] and is a single evaluation artifact, not a sample of independent campaigns.

III. Task and Evaluation Protocol

Let $\mathcal{T}$ be the candidate universe, with $|\mathcal{T}| = V = 835$ technique IDs read from the pinned ATT&CK Enterprise v18.1 STIX bundle. Let the observed ordered sequence be $S = (s_1, \ldots, s_n)$, $n = 140$, with $s_i \in \mathcal{T}$. The first $c = 98$ steps are initial context and the last $m = 42$ are held out, $y_j = s_{c+j}$. At test position j, a method receives only $s_1, \ldots, s_{c+j-1}$ (earlier held-out steps are revealed after they occur: teacher forcing) and returns a ranked list $R_j^{(k)}$ of k distinct candidates from $\mathcal{T}$. Top-k accuracy is

$$A_k = \frac{1}{m} \sum_{j=1}^{m} \mathbb{1}\{y_j \in R_j^{(k)}\}. \tag{1}$$

The run aborts if any observed label is absent from $\mathcal{T}$; held-out labels are never appended to the candidate set. Ties are broken by ascending technique ID, and candidate lists are de-duplicated. A uniform random ranker has expected $A_k = k/835$, i.e., 0.12%, 0.36%, and 0.60% for $k = 1, 3, 5$.

Because the 58 observed labels are a small subset of $\mathcal{T}$, the task is much harder than the closed 58-label formulation of our earlier draft, and scores are not comparable with it. We pose three descriptive questions: (RQ1) do order-aware methods exceed order-agnostic ones on this split? (RQ2) does model complexity help? (RQ3) how much can be concluded from synthetic training data and a single real sequence? No population-level inference is attempted.

IV. Methodology

A. Real Sequence and Source Provenance

The sequence is reconstructed from the public MITRE ATT&CK Evaluations APT29 structured results, pinned to commit `8bee0d3` of the Joystick repository [13], [15]. Flattening the technique-step records and sorting substeps by (major step, branch letter, numeric suborder) gives 140 instances with 140 unique substeps and 58 distinct technique IDs. The McAfee, Microsoft, FireEye, CrowdStrike, and Bitdefender files produced identical ordered (substep, technique) pairs, and each file's Git blob identifier matched the pinned value. This supports reproducibility of the ordering but does not create five samples: the files describe one MITRE-designed emulation flow, not a canonical chronology of all APT29 activity.

B. Candidate Universe

The universe contains every technique and sub-technique ID in the v18.1 Enterprise bundle (SHA-256 recorded in the released manifest): 835 IDs, of which 691 are active, 12 deprecated, and 132 revoked. Deprecated and revoked IDs are retained because the 2019 evaluation uses legacy identifiers. All 58 observed labels are present, but 23 of them are revoked or deprecated in v18.1 (13 of the 42 held-out positions have a revoked target), whereas such IDs are only 17.2% of $\mathcal{T}$. All 58 observed labels are parent-level IDs, while $\mathcal{T}$ also contains sub-techniques. v18.1 was chosen to retain the pre-v19 tactic taxonomy; a later release was not evaluated.

TABLE I
Synthetic corpus statistics per seed (500 sequences each, lengths 20–140). Impact share is the fraction of instances whose technique is tagged Impact.

Seed	Instances	Mean len.	LSTM examples	Impact share
20261011	38,968	77.94	36,060	22.4%
20261012	39,742	79.48	36,834	22.0%
20261013	40,769	81.54	37,861	22.2%

C. Synthetic Corpus

For each seed, 500 sequences with lengths drawn uniformly from 20 to 140 are generated. A walk over the 14 ATT&CK Enterprise tactics in the order Reconnaissance, Resource Development, Initial Access, Execution, Persistence, Privilege Escalation, Defense Evasion, Credential Access, Discovery, Lateral Movement, Collection, Command and Control, Exfiltration, Impact starts at the first tactic and, at each step, advances one tactic ($p = 0.55$), stays (0.25), moves back one or two (0.13), or advances two or three (0.07). The primary condition *reflects* out-of-range positions back into the tactic list. A technique is then sampled uniformly from the visited tactic's pool, which is built from the catalogue's tactic mappings only. A technique can belong to several pools (204 of 835 do), and pool sizes range from 21 (Exfiltration) to 262 (Defense Evasion), so uniform sampling within pools is far from uniform over techniques: techniques in small pools are sampled much more often. Table I gives the realized corpus statistics. The tactic order is a modeling heuristic, not an ATT&CK-mandated sequence [2].

Boundary-rule variant. A variant that *clamps* instead of reflecting (positions saturate at the first and last tactic) is also run as a sensitivity condition. Because forward moves dominate, clamped walks accumulate at the last tactic: 50–51% of instances are Impact-tagged techniques and 66–70% of sequences end in one. None of the 140 observed instances, nor any of the 42 held-out targets, is an Impact-tagged technique; the observed sequence is dominated by Discovery (34 instances), Defense Evasion (32), Execution (19), and Command and Control (15). The originally supplied script used clamping, whereas the project handover described reflection. The reflecting condition matches the handover's seed-level hit counts; the original `artifacts_reflecting_walk/` directory was not supplied, so those prior outputs could not be inspected directly. We report both conditions because the discrepancy materially affects results, and make no claim that either generator is realistic.

D. Models

All methods rank all 835 candidates and report the top five. *Frequency* ranks by synthetic counts plus counts in the permitted prefix. *Co-occurrence* is a local association ranking, not hierarchical clustering: it sums ± 5-position co-occurrence counts for the last six observed techniques. *Markov-1* ranks

TABLE II
PRIMARY CONDITION: TOP-k HITS OUT OF 42 (SEEDS 20261011 / 20261012 / 20261013) AND MEAN ACCURACY ± SAMPLE SD ACROSS SEEDS. UNIFORM RANDOM: 0.12%, 0.36%, 0.60%.

Method	Top-1	Top-3	Top-5
Frequency	0/0/0	0/0/1	1/0/1
	0.0%	0.8±1.4%	1.6±1.4%
Co-occurrence	1/0/0	2/0/1	4/2/2
	0.8±1.4%	2.4±2.4%	6.3±2.8%
Markov-1	0/1/0	3/2/2	5/5/6
	0.8±1.4%	5.6±1.4%	12.7±1.4%
Markov-2 + backoff	2/2/2	3/2/3	5/4/6
	4.8±0.0%	6.3±1.4%	11.9±2.4%
LSTM	0/0/0	0/2/0	2/2/3
	0.0%	1.6±2.8%	5.6±1.4%

TABLE III
SENSITIVITY CONDITION (CLAMPED WALK): TOP-k HITS OUT OF 42 PER SEED AND MEAN ACCURACY. NOT THE PRIMARY RESULT.

Method	Top-1	Top-3	Top-5
Frequency	0/0/0	0/0/0	0/0/0
	0.0%	0.0%	0.0%
Co-occurrence	1/1/2	2/3/3	5/5/5
	3.2%	6.3%	11.9%
Markov-1	1/1/0	4/3/3	6/5/4
	1.6%	7.9%	11.9%
Markov-2 + backoff	3/2/2	5/3/3	6/4/4
	5.6%	8.7%	11.1%
LSTM	0/0/0	0/0/0	0/0/0
	0.0%	0.0%	0.0%

by transition counts from the last technique; *Markov-2 with backoff* uses the last-two-technique context and falls back to first order when that context is unseen. Count-based methods use synthetic counts plus the permitted prefix, updated at every test position. The *LSTM* (32-d embedding, 64 hidden units, dropout 0.2, six-technique window, Adam with learning rate 0.003, batch size 256, 35 epochs, cross-entropy) is trained once per seed on windows from the synthetic sequences plus the first 98 observed steps only (92 observed windows against roughly 37,000 synthetic ones); at test time it conditions on the last six steps of the permitted prefix. These hyperparameters were fixed before the run and were not tuned. Count-based methods and the LSTM therefore differ in how they absorb the observed prefix, which should be kept in mind when comparing them.

E. Reproducibility Controls

Each run records script SHA-256 hashes, input SHA-256 hashes, package versions, seeds, and settings in a manifest, writes the generated corpora, and exports all 42 ranked lists per seed. Output directories must be new, so earlier artifacts are not overwritten. An independent script re-derived the sequence split, catalogue coverage (directly from the STIX bundle), ranking validity (five unique in-catalogue candidates in every ranked list of both conditions), and every hit count, and agreed with the recorded metrics in all cases. The controlled four-condition suite was executed under one recorded runtime and independently reconciled from CSV outputs. The primary condition matches the seed-level hit counts preserved in the handover, although the original run directory was unavailable for direct comparison. Cross-platform bit identity has not been established.

V. RESULTS

Table II reports hit counts per seed and mean accuracy across the three seeds for the primary (reflecting) condition. Seeds change the synthetic corpus and LSTM initialization; the 42 evaluated positions are identical across seeds, so the three runs are not independent observations of campaign behavior.

Three descriptive observations follow. First, the Markov models are the only methods with consistent top-5 hits above the 0.6% uniform reference (about 0.25 expected hits): 4–6 of 42 for both, and Markov-2 is the only method with a non-zero top-1 count in every seed (2 of 42). Co-occurrence and the LSTM register 2–4 and 2–3 top-5 hits, and global frequency 0–1. Second, differences among the three better methods are one to three hits per cell and comparable to the seed-to-seed spread, so no reliable ranking between Markov-1 and Markov-2 (or between either and co-occurrence at top-5) is supported. Third, absolute accuracy is low: the union of all five methods' top-5 lists contains the true technique at only 9, 9, and 11 of the 42 positions in the three seeds.

Of the 42 held-out positions, 11 have a target label absent from the 98-step context (8 distinct labels; 8 of these positions are first occurrences in the whole sequence and 3 repeat an earlier held-out label), and 31 targets appeared in the context. Methods that depend on prior observations of the label cannot rank a first occurrence from the observed prefix; they can do so only through synthetic counts. We did not break accuracy down further because the numbers of positions per stratum are too small for a meaningful comparison.

Because each cell is a count out of 42 dependent positions from one emulation, we do not report confidence intervals or significance tests: an independent-binomial interval would understate dependence, and a block bootstrap needs more sequences. The standard deviations in Table II describe sensitivity to seeds only.

A. Sensitivity to the Generator Boundary Rule

Table III gives the same measurements when the tactic walk clamps instead of reflecting. Everything else (seeds, catalogue, models, hyperparameters, split) is unchanged. Mean top-5 accuracy becomes 11.9% for co-occurrence, 11.9% for Markov-1, 11.1% for Markov-2, and 0% for frequency and the LSTM, with no hits at any k for the last two. Co-occurrence rises from 6.3% to 11.9% at top-5 while the LSTM falls from 5.6% to 0%. The top-five frequency candidates are techniques tagged only Exfiltration (the smallest pool) in the reflecting condition and only Impact in the clamped condition, with no hits by this baseline in the clamped condition. We did not tune either generator against the held-out scores; the point is that a seemingly minor, unvalidated generator choice changes the ordering of methods, which limits what any ranking from synthetic-trained models can mean.

B. Sensitivity to Tactic Sampling and Synthetic Status

Two additional policy checks separate the effects of tactic transitions and catalogue-status filtering. The uniform-tactic condition samples a tactic independently and uniformly at every synthetic position, rather than following the ordered walk. The active-only condition retains the reflecting walk but samples synthetic techniques only from catalogue objects not marked deprecated or revoked. In both conditions, the evaluation vocabulary remains the complete independent 835-ID catalogue, and the observed initial context is unchanged. These are prespecified stress tests of the generator, not claims that either policy is more realistic.

The additional policies reinforce the sensitivity caveat. Uniform tactic sampling raises co-occurrence Top-5 from the primary 6.3% to 10.3%, while the LSTM falls from 5.6% to 0.8%. Restricting synthetic choices to active techniques lowers co-occurrence Top-5 to 3.2%; Markov-2 remains at 11.9%, and the LSTM is 4.0%. These results do not identify a superior generator. They show that rankings vary with synthetic assumptions and that status filtering changes training counts even when the ranking universe remains fixed.

C. Earlier Historical Figures (Not Reproduced)

An earlier draft reported the scores in Table V for a closed 58-label vocabulary derived from the evaluation sequence itself, with six method configurations including a hierarchical co-occurrence clustering method. Its code, corpus, and per-position predictions are unavailable, so these values are unverified, were not recomputed, and are shown only for provenance. They are not comparable with Table II, which uses a different candidate universe, protocol, and implementation, and they should not be cited as evidence.

VI. Discussion

RQ1: the Markov models, which condition on the immediately preceding technique(s), are the strongest in the primary condition and the only ones with non-zero top-1 hits in every seed, which is consistent with, but does not demonstrate, a signal from the preceding techniques beyond global prevalence. RQ2: added complexity did not help. The LSTM is the most complex model; it is trained mostly on synthetic windows (92 observed windows against roughly 37,000 synthetic ones), so it is likely to reproduce the synthetic transition structure, but we did not test this explanation, and its low scores may also reflect training choices that were deliberately not tuned. It also went from 5.6% to 0% top-5 when only the boundary rule changed, which suggests that its behavior is dominated by the synthetic distribution. RQ3: very little can be concluded. Synthetic volume is not independent real-world evidence, sampling within tactic pools can impose a tactic progression without capturing real transition frequencies, branching, or campaign specifics, and the ordering of co-occurrence, frequency, and LSTM changes with a boundary rule that has no empirical justification.

The 835-way universe contains many revoked and deprecated IDs, and because 13 of 42 targets are themselves revoked legacy IDs, the evaluation partly measures the ability to rank legacy identifiers that a current ATT&CK-based detector would map to successor techniques. A mapping from legacy to current IDs, or a different candidate policy, would change the task and was not evaluated.

A free-running rollout should be reported separately from teacher-forced prediction, since in a rollout errors compound. We did not run one.

VII. Limitations and Required Validation

(1) The evaluation has one reconstructed emulation sequence; five vendor files are one flow. (2) The 42 positions are sequentially dependent and 11 targets are unseen in the context; seeds do not create independent campaigns. (3) The synthetic generator is a heuristic, is not validated as an attack simulator, and its results are sensitive to the boundary rule (Table III), tactic-selection policy, and synthetic technique-status filter (Table IV); we did not search over generator variants against held-out scores, and no tested variant is claimed as realistic. (4) Neither the baselines nor the LSTM were tuned, so the weak LSTM results should not be read as a bound on neural methods. (5) The co-occurrence baseline is local association ranking and is not the historical clustering method. (6) The catalogue policy (v18.1 with revoked and deprecated IDs, parent-level targets) affects the task, and other releases were not evaluated. (7) No uncertainty estimate that respects dependence is provided. Claims beyond this pilot would need multiple independently sourced ordered sequences, with leave-one-campaign-out evaluation, a validated or ablated generator, and a free-running comparison.

VIII. Reproducibility and Data Availability

Source hashes, pinned revisions, scripts, generated corpora, per-position rankings, per-seed metrics, and independent reconciliation reports are released with the anonymized supplementary material. The controlled four-condition suite was rerun with Python 3.13.5, NumPy 2.3.5, and PyTorch 2.10.0+cpu, using one CPU intra-op thread; manifests record the runtime and every generator policy. Repeated outputs in the original controlled runs were byte-identical, but bit identity across operating systems and library versions has not been established. The code is a new clean-room implementation and is not a recreation of the lost historical experiment.

IX. Conclusion

On one APT29 emulation sequence and an independent 835-ID ATT&CK v18.1 candidate universe, first- and second-order Markov models placed the next technique in their top five for 4–6 of 42 held-out positions, while global frequency, co-occurrence, and an LSTM trained mostly on synthetic data did less well; changing only the generator's boundary rule reorders these methods. Given one sequence, 42 dependent positions, and an unvalidated generator, these are descriptive pilot results and not evidence of cross-campaign or operational predictive capability. The released artifacts allow every number to be audited, and the next steps are independent sequences, a validated generator, and held-out-campaign evaluation.

TABLE IV

ADDITIONAL SYNTHETIC-GENERATION SENSITIVITY: MEAN TOP-k ACCURACY (%) OVER THREE SEEDS ON THE SAME 42 HELD-OUT POSITIONS. SEED RUNS ARE NOT INDEPENDENT CAMPAIGNS.

	Uniform tactic sampling			Active-only synthetic pool		
Method	Top-1	Top-3	Top-5	Top-1	Top-3	Top-5
Frequency	0.0	0.8	0.8	0.8	0.8	0.8
Co-occurrence	5.6	7.9	10.3	0.0	2.4	3.2
Markov-1	0.8	4.0	11.1	2.4	10.3	10.3
Markov-2 + backoff	5.6	6.3	11.9	7.1	11.9	11.9
LSTM	0.0	0.0	0.8	1.6	3.2	4.0

TABLE V

HISTORICAL, UNVERIFIED, NOT REPRODUCED (58-LABEL TASK; DO NOT COMPARE WITH TABLE II).

Method (historical label)	Top-1	Top-3	Top-5
Global frequency	0.024	0.048	0.071
Co-occurrence clustering	0.024	0.048	0.071
Markov, 2nd-order	0.000	0.071	0.143
Markov, 1st-order	0.000	0.071	0.167
LSTM, pretrain + fine-tune	0.024	0.071	0.095
LSTM, mixed training	0.000	0.095	0.119

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